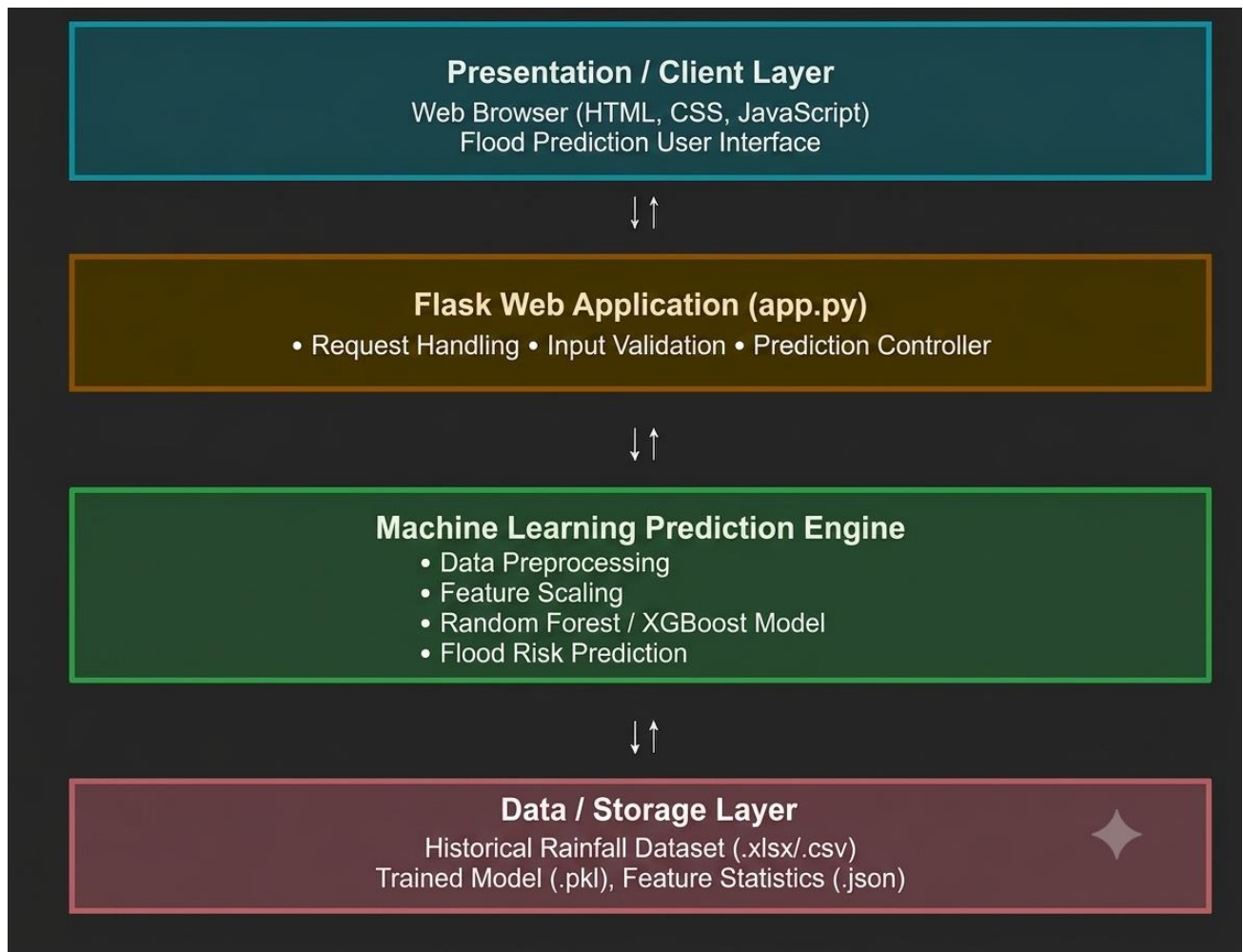


Solution Architecture

Date	10 July 2026
Team ID	XXXXXX
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level architectural representation of the **Rising Waters: A Machine Learning Approach to Flood Prediction** system. The architecture illustrates how users interact with the Flask web application, how weather and rainfall data are processed by the machine learning model, and how prediction results are generated using historical datasets and displayed to users.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
<i>Presentation Layer</i>	<i>Provides a user-friendly web interface for entering rainfall and weather parameters and displaying flood prediction results.</i>	<i>HTML5, CSS3, JavaScript, Bootstrap</i>
<i>Application Layer (Flask Server)</i>	<i>Handles HTTP requests, validates user input, loads the trained machine learning model, performs predictions, and returns results to the user.</i>	<i>Python, Flask</i>
<i>Machine Learning Prediction Engine</i>	<i>Processes input data, applies preprocessing and feature scaling, executes the trained classification model, and predicts flood risk.</i>	<i>Scikit-learn, Random Forest, Decision Tree, KNN, XGBoost, Pandas, NumPy</i>
<i>Data Layer</i>	<i>Stores historical rainfall datasets, trained machine learning models, preprocessing objects, and configuration files required for prediction.</i>	<i>Excel (.xlsx), CSV, Pickle (.pkl), JSON</i>